

AKSHITHA SURABHI

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SUMMARY

- Data Engineer experienced in designing, building, and operating cloud-native data platforms across financial services and healthcare domains.
- Strong expertise in developing ETL/ELT pipelines, real-time streaming systems, and analytics-ready data models for enterprise use cases.
- Hands-on experience with Python (PySpark), Spark, Databricks, Airflow, Kafka, and SQL for large-scale data processing.
- Proficient across AWS, Azure, and GCP, including data lakes, data warehouses, and cloud-native orchestration services.
- Experienced in implementing lakehouse architectures (Bronze, Silver, Gold) with performance optimization and data quality enforcement.
- Skilled in Infrastructure as Code (Terraform), CI/CD automation, and DevOps practices for reliable and repeatable deployments.
- Background in real-time and event-driven architectures using Kafka, Kinesis, and Azure Event Hubs.
- Strong focus on data security, governance, and compliance, supporting regulated enterprise environments.

TECHNICAL SKILLS

Cloud Platforms: Microsoft Azure (ADF, Synapse, Azure SQL DB, Event Hub, AKS, Data Lake Gen2, DevOps), AWS (Lambda, Glue, EMR, S3, Redshift, DynamoDB, Kinesis, EC2, CloudFormation, CloudWatch), GCP

Programming & Scripting: Python, SQL, R, PL/SQL, Shell, PowerShell, Scala, REST APIs

DevOps & Infrastructure as Code (IaC): Terraform, Azure DevOps (YAML), Jenkins, Git, Docker, CI/CD Automation

Data Warehousing & Visualization: Redshift, Snowflake, Azure Synapse, Power BI, Tableau, Excel (Power Pivot, Power Query)

Security & Monitoring: Azure Key Vault, AWS Secrets Manager, IAM Policies, CloudWatch, Data dog, Grafana

Data & Streaming: Spark, Databricks, Kafka (Apache kafka), Airflow, Event Hub, Kinesis

Databases: SQL Server, Oracle, PostgreSQL, MySQL, Dynamo DB, MongoDB, Snowflake

Machine Learning: Scikit-learn, TensorFlow, Pandas, NumPy, Feature Engineering, Model Deployment, Regression, Hypothesis Testing

Formats: Parquet, Avro, JSON, CSV

Architectures: Batch & Streaming Data Pipelines, Lakehouse (Bronze/Silver/Gold), Event-Driven Architectures, Cloud Data Warehousing

EXPERIENCE

EDWARD JONES	USA
Data Engineer - Financial Services	June 2024 - Present
• Designed and supported secure, enterprise-scale data platforms using AWS and GCP to enable financial analytics, reporting, and advisory use cases. • Built and orchestrated data ingestion and transformation pipelines using Apache Airflow (GCP Composer) integrating cloud storage, data warehouses, and serverless services. • Implemented real-time streaming pipelines for market and transactional data using AWS Kinesis and Lambda, delivering near real-time insights to downstream systems. • Developed PySpark and Scala-based data processing pipelines to handle high-volume datasets with strong focus on data accuracy, lineage, and auditability. • Automated cloud infrastructure provisioning using Terraform, enforcing IAM-based security, access controls, and secrets management aligned with compliance standards.	
<i>Industry Focus: Wealth Management, Financial Analytics & Reporting, Regulatory & Compliance Data Platforms</i>	
CIGNA HEALTHCARE	Hyderabad, India
Data Engineer - Healthcare & Insurance	Jul 2021 - Nov 2022
• Designed and implemented Azure Data Factory (ADF) pipelines to ingest, validate, and transform healthcare and insurance data into Azure Data Lake Gen2 and Azure Synapse for enterprise analytics. • Developed Databricks PySpark workflows to support claims data processing, member analytics, and machine-learning feature engineering. • Built near real-time ingestion pipelines using Azure Event Hubs and Azure Stream Analytics, persisting processed data into Cosmos DB and Synapse. • Ensured HIPAA-compliant data handling by implementing secure data access patterns, encryption at rest and in transit, and role-based access controls (RBAC). • Automated build and deployment processes using GitHub Actions and Azure DevOps CI/CD pipelines, improving deployment consistency and reliability. • Optimized Spark job performance and cluster utilization through tuning and resource right-sizing, achieving approximately 35% reduction in processing costs.	
<i>Industry Focus: Healthcare Analytics / Insurance & Claims Processing / HIPAA-Compliant Data Platforms</i>	
HSBC	Hyderabad, India
Data Engineer - Banking & Risk Analytics	May 2019 - Jun 2021
• Designed and maintained batch and real-time data pipelines on AWS to support core banking operations, risk analytics, and regulatory reporting. • Developed high-performance Spark (Scala) and PySpark pipelines to process large volumes of transactional and customer data, loading curated datasets into Snowflake and Amazon Redshift. • Orchestrated end-to-end workflows using Apache Airflow, enforcing data quality checks, SLA monitoring, and audit-ready processing. • Implemented event-driven architectures using AWS Lambda, integrating S3, DynamoDB, and API Gateway to automate banking workflows and data movement.	

PAPER PUBLICATIONS

Comparison of COVID-19 diagnosis by CNN model and Resnet using chest X-ray Images ([LINK](#))

IEEE | International Conference on Sustainable Communication Networks and Applications (Nov 15–17, 2023)

- Designed and implemented an end-to-end machine learning pipeline to classify chest X-ray images as COVID-19 or non-COVID using a custom CNN and transfer-learning with ResNet.
- Built a production-style data preprocessing and feature engineering workflow, including image normalization, augmentation (rotation, flipping, contrast adjustment), and handling class imbalance to improve model robustness.
- Evaluated models using clinically relevant metrics such as accuracy (~94%), AUC, sensitivity, and specificity, emphasizing reliability over accuracy alone.
- Developed a Python-based web application to serve model predictions, demonstrating model deployment and usability for real-world scenarios.
- Published findings in IEEE, highlighting the importance of data quality, validation, and reproducibility in applied machine learning systems.

Tools & Technologies: Python, TensorFlow, Keras, OpenCV, NumPy, Pandas, scikit-learn

EDUCATION

UNIVERSITY OF HOUSTON

Master of Science in Engineering Data Science

CERTIFICATIONS & AWARDS

- Microsoft Certified: Azure Data Engineer Associate (2025)
- Databricks Academy Accreditation: Databricks for Data Engineering (2024)
- NPTEL: Microprocessors and Microcontrollers